

知识图谱对大模型在企业 SQL 库上问答准确率的作用：一个基准 A Benchmark to Understand the Role of Knowledge Graphs on Large Language Model's Accuracy for Question Answering on Enterprise SQL Databases

核心内容

Q: 这篇论文到底测了什么？ A: 同一批问题、同一个 GPT-4、同一份数据，只换一件事——给模型看的是原始 SQL 建表语句，还是一层业务本体。结果是准确率 16.7% 对 54.2%，差三倍。作者的结论一句话：**要提高大模型问数据的准确率，得先投资在业务语境上。**

Q: 为什么要另做一个基准，现成的 Spider、WikiSQL 不够用吗？ A: 作者认为那些基准和企业现场对不上，三条：一、它们的库结构太简单，企业库动辄几百张表；二、它们不问企业真正关心的问题——报表、指标、KPI；三、**最关键的一条，它们完全没有业务语境层**（元数据、映射、口径、本体）。没有这三样，大模型在企业里就是在真空中答题。

Q: 这个基准长什么样？ A: 三块。一是企业 SQL 表结构，用的是对象管理组织发布的财产与意外险数据模型，物理模型 199 张表、243 个外键，实验取了 13 张的子集。二是 43 道自然语言问题，按两个维度分成四个象限：问题本身复杂不复杂（日常报表 对 指标 KPI）、答它要连几张表（4 张以内 对 4 张以上）。三是语境层，包括一份描述业务概念、属性、关系的本体，以及从 SQL 表结构到本体的映射规则。

Q: 结果里最狠的是哪一条？ A: **表结构一复杂，不带本体的直接问法准确率是零，不是低，是零。** 两个高表结构复杂度的象限，SQL 路线一道都没答对；同样的问题走知识图谱，还能到 35.7% 和 38.7%。作者顺手给了一条经验线：**一道题要连 5 张以上的表，SQL 路线的准确率就掉到零。**

Q: 两条路线错的方式一样吗？ A: 不一样，这是全文最值得记的一处。**SQL 路线的错是「编」**——编不存在的列名、编数据库里没有的取值当筛选条件、编看着合理其实不对的表连接。**知识图谱路线的错是「走错路」**——起点和终点都找对了，中间那条属性路径接错，或者把关系方向搞反。作者说得很直白：**前者是幻觉，后者是路径不一致。** 而且他们手工翻了所有生成的 SPARQL，一个编造的类或属性都没见到。

Q: 作者自己承认了哪些没做到？ A: 不少，而且写得很诚实。只用了 13 张表的子集；数据是手工造的、每张表就几行；表结构是第三范式，没试过星型、雪花、宽表；问题里没有时间范围和数值区间的筛选；只测了 GPT-4 一个模型；提示词是最简单的零样本，没试少样本和思维链；映射全是一一对一的表列映射，没有条件映射、拼接、公式这些复杂形态。

背景补充 / 点评

先说这三个作者是谁，这决定了这篇东西的分量。Dean Allemang 是《Semantic Web for the Working Ontologist》的作者，那本书是语义网领域被引最多的实践教材之一。Juan Sequeda 做 R2RML（关系库到 RDF 的映射标准）相关工作多年。两人都在 data.world，那是一家做数据目录和知识图谱的公司。所以这篇有明显的立场——他们卖的就是知识图谱，这一点读的时候要放在心上。

但立场不影响这个实验的价值，因为它的设计是干净的。两条路线用的是同一个模型、同一份数据、同一套评分、几乎同样的提示词模板，只差「给它看 DDL」还是「给它看本体」。而且他们把全部数据、本体、映射、问题、参考查询都发到了 GitHub，任何人可以拿自己公司的表结构重跑一遍。这比结论本身更有用。

评分办法也值得抄。他们没有比生成的查询语句像不像，而是比执行结果——两个查询跑出来的表内容一致就算对，列的顺序、列名、行的顺序都不管。这一条绕开了「同一个问题有多种正确写法」这个老问题。更细的是他们承认大模型是不确定的，所以每道题跑很多次，一道题的分数是「跑对的次数除以总次数」，不是一次定生死。

我认为洞察最深的是「部分正确」那一节。他们手工翻查询时发现有一类答案是「对但不全」——返回的列都对，只是少了一列；或者返回了内部主键而不是业务编号。作者说：从实际使用看，这种答案用户可能是能接受的，因为用户可以追问一句补上。但怎么定义、怎么给分，他们没有答案，明确留成了开放问题。这一处的诚实程度在论文里不多见。

留白最大的地方是「为什么 SPARQL 不产生幻觉」。他们自己也说这很意外，日常用 ChatGPT 问同类问题时是会编属性的。他们猜可能是提示词里那句「只返回查询语句，不要解释」起的作用，但同一句话 SQL 那边也有，为什么只在一边起效，他们说不清。这个问题悬着。

对我本人的启示，三条。

第一条最直接：我们那份消融实验的数字，现在有外部对照了。我们自己测的是同模型重跑 14/15 对 10/15 对 7/15，一直只有自家一组数。这篇给的是 54.2% 对 16.7%，方向一致、量级更极端。更要紧的是它给了一个我们没测过的切面——按表结构复杂度分层。我们的验证问题大多落在单表或少数几张表上，「一道题要连 5 张以上表时准确率掉到零」这条线，我们从来没试过。这是个该补的口子。

第二条是它的失效分类。SQL 路线编列名、编取值、编连接——这三样正好对应我们那条「LLM 不写 SQL、不产生数字」的硬纪律。我们是靠 metric-registry 的 DSL 声明经编译器生成 SQL，从结构上把这三种幻觉堵死了。这篇论文相当于给那条纪律提供了公开的量化证据：不堵，准确率就是 16.7%。

第三条是个警告，冲着我们的语义层来的。知识图谱路线的错是「路径走错、方向搞反」。我们的 rbox.yaml 定义的关系正是这个东西。论文说大模型知道起点和终点，就是中间的路接不对——那意味着关系定义得越绕、跳数越多，出错概率越大。这一条对我们的启示是：join 路径要短、要显式登记，别指望模型自己拼。我们现在的做法是关系必须在 rbox 里登记、编译器按登记的路径生成，方向是对的，但这篇提醒了为什么这件事不能松。

还有一条不是启示是提醒：他们把「知识工程方法论」列进了研究议程，原话是现有的本体与知识工程方法论是在数据量小的年代定的，而且是纯手工流程，怎么改造它们、怎么把大模型也用进来降低建本体的成本，是

个开放问题。这正是我们这几周在做的事——而且我们是纯手工在做。

文中金句

"Without these vital components, LLMs for enterprise question answering risk being disconnected from the reality of enterprise data, leading to hallucinations and uncontrolled outcomes." 缺了这些关键部件，用在企业问答上的大模型就有和企业数据的真实情况脱节的风险，导致幻觉和失控的结果。

"The results also lead us to understand when SQL starts to fail. When a question requires more than 5 tables to provide then answer, the accuracy drops to zero." 这些结果还让我们看清了 SQL 路线是从哪里开始失效的：当一个问题需要连 5 张以上的表才能回答时，准确率掉到零。

"The inaccuracy of SQL queries are based on hallucination while the inaccuracy of SPARQL queries are based on path inconsistency." SQL 查询的错误源于幻觉，而 SPARQL 查询的错误源于路径不一致。

"We argue that this context needs to be treated as a first class citizen, thus, at minimum managed in a metadata management system (i.e. data catalog) and ideally on a knowledge graph architecture. Otherwise, the crucial context that provides accuracy would be managed in an ad-hoc manner." 我们主张这层语境必须被当作一等公民对待，至少要在元数据管理系统（也就是数据目录）里管起来，理想情况下管在知识图谱架构上。否则，这份提供准确率的关键语境就会被以临时凑合的方式管理。

全文翻译

摘要

Enterprise applications of Large Language Models (LLMs) hold promise for question answering on enterprise SQL databases. However, the extent to which LLMs can accurately respond to enterprise questions in such databases remains unclear, given the absence of suitable Text-to-SQL benchmarks tailored to enterprise settings. Additionally, the potential of Knowledge Graphs (KGs) to enhance LLM-based question answering by providing business context is not well understood.

大语言模型在企业中的应用，为在企业 SQL 数据库上做问答带来了前景。然而，由于缺少针对企业场景的 Text-to-SQL 基准，大模型究竟能在多大程度上准确回答这类数据库上的企业问题，一直不清楚。此外，知识图谱通过提供业务语境来增强基于大模型的问答，这个潜力也还没有被充分理解。

This study aims to evaluate the accuracy of LLM-powered question answering systems in the context of enterprise questions and SQL databases, while also exploring the role of knowledge graphs in improving accuracy. To achieve this, we introduce a benchmark comprising an enterprise SQL schema in the insurance domain, a range of enterprise queries encompassing reporting to metrics, and a contextual layer incorporating an ontology and mappings that define a knowledge graph.

本研究旨在评估大模型驱动的问答系统在企业问题与 SQL 数据库这一语境下的准确率，同时探究知识图谱在提升准确率上的作用。为此，我们引入一个基准，它由三部分构成：一份保险领域的企业 SQL 表结构；一组覆盖从报表到指标的企业查询；以及一个语境层，其中包含一份本体和一套映射，两者共同定义了一个知识图谱。

Our primary finding reveals that question answering using GPT-4, with zero-shot prompts directly on SQL databases, achieves an accuracy of 16%. Notably, this accuracy increases to 54% when questions are posed over a Knowledge Graph representation of the enterprise SQL database. Therefore, investing in Knowledge Graph provides higher accuracy for LLM powered question answering systems.

我们的主要发现是：用 GPT-4 以零样本提示直接在 SQL 数据库上做问答，准确率是 16%。值得注意的是，当同样的问题是在该企业 SQL 数据库的知识图谱表示之上提出时，这个准确率上升到 54%。因此，[投资于知识图谱能让大模型驱动的问答系统获得更高的准确率](#)。

一、引言：为什么现有基准不够用

Question answering, the ability to interact with data using natural language questions and obtaining accurate results, has been a long-standing challenge in computer science dating back to the 1960s. The field has advanced throughout the past decades, through Text-to-SQL approaches, as a means of facilitating chatting with the data that is stored in SQL databases. With the rise of Generative AI and Large Language Models (LLMs), the interest continues to increase.

问答——也就是用自然语言问题与数据交互并获得准确结果的能力——是计算机科学里一个由来已久的挑战，可以追溯到 1960 年代。过去几十年里，这个领域通过 Text-to-SQL 路线不断推进，作为与存储在 SQL 数据库中的数据对话的一种手段。随着生成式 AI 与大语言模型的兴起，人们对它的兴趣仍在增长。

While question answering systems have shown remarkable performance in several Text-to-SQL benchmarks, such as Spider, WikiSQL, KaggleDBQA, their implications relating to enterprise SQL databases remain relatively obscure. We argue that existing Question Answering and Text-to-SQL benchmarks, although valuable, are often misaligned with real-world enterprise settings:

尽管问答系统在若干 Text-to-SQL 基准上（如 Spider、WikiSQL、KaggleDBQA）表现亮眼，但它们对企业 SQL 数据库意味着什么，仍然相当模糊。我们认为，现有的问答与 Text-to-SQL 基准虽有价值，却常常与真实的企业场景对不上：

1. these benchmarks typically overlook complex database schemas representing enterprise domains, which likely comprise hundreds of tables,
2. they also often disregard questions that are crucial for operational and strategic planning in an enterprise, including questions related to business reporting, metrics, and key performance indicators (KPIs), and
3. a critical missing link is the absence of a business context layer - metadata, mappings, transformations, ontologies, that provides business semantics and knowledge about the enterprise.

一、这些基准通常忽略了代表企业领域的复杂库结构，那种结构很可能包含数百张表；二、它们也常常不考虑那些对企业运营与战略规划至关重要的问题，包括与业务报表、指标和关键绩效指标相关的问题；三、**一个关键的缺失环节是没有业务语境层**——元数据、映射、转换、本体，这些东西提供关于这家企业的业务语义与知识。

Without these vital components, LLMs for enterprise question answering risk being disconnected from the reality of enterprise data, leading to hallucinations and uncontrolled outcomes. Their inability to provide explainable results can significantly impede their trustworthiness and adoption.

缺了这些关键部件，用在企业问答上的大模型就有和企业数据的真实情况脱节的风险，导致幻觉和失控的结果。它们无法给出可解释的结果，会严重妨碍其可信度与被采纳的程度。

Knowledge Graphs (KGs) have been identified as a promising solution to fill the business context gaps in order to reduce hallucinations, thus enhancing the accuracy of LLMs. Similarly, from an industry perspective, Gartner states, "Knowledge graphs provide the perfect complement to LLM-based solutions where high thresholds of accuracy and correctness need to be attained."

知识图谱已被视为填补业务语境缺口、从而减少幻觉、进而提升大模型准确率的一条有希望的路径。同样，从产业界的视角看，Gartner 表示：「在需要达到高准确率与正确性门槛的场合，知识图谱为基于大模型的方案提供了完美的补充。」

二、研究问题与假设

RQ1: To what extent Large Language Models (LLMs) can accurately answer enterprise natural language questions over enterprise SQL databases. RQ2: To what extent Knowledge Graphs can improve the accuracy of Large Language Models (LLMs) to answer enterprise natural language questions over enterprise SQL databases.

研究问题一：大语言模型在企业 SQL 数据库上，能在多大程度上准确回答企业自然语言问题。**研究问题二：**知识图谱能在多大程度上提升大语言模型回答这类问题的准确率。

The hypothesis is the following: An LLM powered question answering system that answers a natural language question over a knowledge graph representation of the SQL database returns more accurate results than a LLM powered question answering system that answers a natural language question over the SQL database without a knowledge graph.

假设如下：一个在 SQL 数据库的知识图谱表示之上回答自然语言问题的、大模型驱动的问答系统，比一个在没有知识图谱的 SQL 数据库上直接回答自然语言问题的同类系统，返回更准确的结果。

三、基准框架

3.1 企业 SQL 表结构

The enterprise SQL schema used in the benchmark comes from the P&C Data Model for Property And Casualty Insurance, a standard model created by Object Management Group (OMG), a standards development organization. The main entities in this Property and Casualty Data Model consist of: Account, Activity, Agreement, Claim, Communication, Coverage, Event, Geographic Location, Insurable Object, Location Address, Money, Party, Policy, Policy Coverage Detail, Policy Deductible, Policy Limit, Product.

基准所用的企业 SQL 表结构来自财产与意外险数据模型，那是标准制定组织「对象管理组织」创建的一个标准模型。该模型中的主要实体包括：账户、活动、协议、理赔、沟通、承保范围、事件、地理位置、可保标的、位置地址、货币、当事方、保单、保单承保明细、保单免赔额、保单限额、产品。

The physical data model consists of 199 tables. Each table has a primary key. There are a total of 243 foreign keys. For this current version of the benchmark, we consider a subset of the SQL schema, 13 tables.

物理数据模型包含 199 张表。每张表都有主键。外键共计 243 个。在当前这一版基准中，我们只取该表结构的一个子集，13 张表。

3.2 企业问题：两个维度切出四个象限

The questions are classified on a spectrum of low to high complexity: Low question complexity: Pertains to business reporting use cases, aimed at facilitating daily business operations. From a technical standpoint, these questions are translated into SELECT-FROM SQL queries. High question complexity: Arises in the context of Metrics and Key Performance Indicators (KPIs) within an organization. From a technical standpoint, these questions are translated to SQL queries involving aggregations and mathematical functions.

问题按复杂度从低到高分类。**低问题复杂度**：属于业务报表类用途，服务于日常业务运营；从技术角度看，这类问题会被翻译成「查询-来自」形式的 SQL。**高问题复杂度**：出现在组织内部的指标与关键绩效指标语境中；从技术角度看，这类问题会被翻译成涉及聚合与数学函数的 SQL。

Questions also depend on the number of tables required to provide an answer. Low schema complexity: Small number of tables (i.e. 0 - 4), denormalized schema. High schema complexity: Larger number of tables (> 4), normalized schema, many-to many join tables, etc.

问题还取决于回答它需要用到多少张表。**低表结构复杂度**：表数量少（0 到 4 张），反范式的结构。**高表结构复杂度**：表数量较多（超过 4 张），规范化的结构，多对多的连接表，等等。

3.3 语境层：本体加映射

The context layer consists of two parts: Ontology: Business Concepts, Attributes, and Relationships that describe the insurance domain. Mapping: transformation rules from the source SQL schema to the corresponding Business Concepts, Attributes, and Relationships in the target ontology.

语境层由两部分构成。**本体**：描述保险领域的业务概念、属性与关系。**映射**：从源 SQL 表结构到目标本体中相应业务概念、属性与关系的转换规则。

For this current version of the benchmark, the context layer is provided in machine readable as RDF: ontology in OWL and mapping in R2RML. The OWL ontology and R2RML mappings can be used to create the Knowledge Graph either in a virtualized or materialized way.

在当前这一版基准中，语境层以机器可读的 RDF 提供：本体用 OWL，映射用 R2RML。这份 OWL 本体与 R2RML 映射，可以用虚拟化或物化两种方式来构建知识图谱。

3.4 评分：只比执行结果，不比语句

Execution Accuracy (EA) We follow the metric of Execution Accuracy (EA) from the Spider benchmark. An execution is accurate if the result of the query matches the answer for the query. Note that the order or the labels of the columns are not taken in account for accuracy.

执行准确率：我们沿用 Spider 基准的执行准确率指标。如果一次查询的结果与该问题的标准答案一致，这次执行就算准确。注意，列的顺序和列的标签不计入准确性判断。

Overall Execution Accuracy (OEA) Given the non-deterministic nature of LLMs, there is no guarantee that given an input question, the generated query will always be the same thus providing the same answer. Therefore, every question has a Overall Execution Accuracy (OEA) score which is calculated as (# of EA)/Total Number of runs.

总体执行准确率：鉴于大模型的不确定性，无法保证同一个输入问题每次生成的查询都相同、从而给出相同答案。因此每道题都有一个总体执行准确率分数，算法是「执行准确的次数除以总运行次数」。

四、实验设置

The question answering system we evaluated was a zero-shot prompt to GPT-4, that is instructed to generate a query, which is run against the database. The resulting response is compared to the response given by the reference query. The particular parameters to the OpenAI API are as follows: max_tokens = 2048, n = 1, temperature = 0.3.

我们评估的问答系统是给 GPT-4 一个零样本提示，让它生成一条查询，这条查询会在数据库上执行，其结果与参考查询给出的结果作比较。OpenAI 接口的具体参数是：最大 token 数 2048，n 为 1，温度 0.3。

SQL Zero-shot Prompt: "Given the database described by the following DDL: INSERT SQL DDL. Write a SQL query that answers the following question. Do not explain the query. return just the query, so it can be run verbatim from your response. Here's the question: INSERT QUESTION"

SQL 零样本提示词：「给定由以下 DDL 描述的数据库：插入 SQL DDL。写一条能回答下面这个问题的 SQL 查询。不要解释这条查询。只返回查询本身，以便能从你的回答里原样运行。问题是：插入问题。」

SPARQL Zero-shot Prompt: "Given the OWL model described in the following TTL file: INSERT OWL ontology. Write a SPARQL query that answers the question. The data for your query is available in a SERVICE identified by <mapped>. Do not explain the query. return just the query, so it can be run verbatim from your response."

SPARQL 零样本提示词：「给定由以下 TTL 文件描述的 OWL 模型：插入 OWL 本体。写一条能回答该问题的 SPARQL 查询。你查询所需的数据在一个标识为 mapped 的服务中。不要解释这条查询。只返回查询本身，以便能从你的回答里原样运行。」

We kept the prompt simple for this experiment, because we wanted to focus on the ability of the contextual information (the DDL in the case of SQL) to provide necessary information for the formation of the query.

这次实验我们把提示词保持得很简单，因为我们想聚焦的是**语境信息本身**（在 SQL 那一路就是 DDL）为构造查询提供必要信息的能力。

五、结果

5.1 总体

By grouping all the questions in the benchmark, SQL achieves an AOEA of 16.7%. In comparison, SPARQL achieves an average OEA of 54.2%. Therefore, overall SPARQL accuracy was 3x the SQL accuracy.

把基准里所有问题合在一起看，SQL 路线的平均总体执行准确率是 16.7%。相比之下，SPARQL 路线是 54.2%。因此**总体上 SPARQL 的准确率是 SQL 的三倍。**

5.2 分象限

Low Question/Low Schema: SQL achieves an AOEA of 25.5%. In comparison, SPARQL achieves an AOEA of 71.1%. The SPARQL accuracy is 2.8X the SQL accuracy. High Question/Low Schema: SQL achieves an AOEA of 37.4%. In comparison, SPARQL achieves an AOEA of 66.9%. The SPARQL accuracy is 1.8X the SQL accuracy. Low Question/High Schema: SQL was not able to answer any question accurately. In comparison, SPARQL achieves an AOEA of 35.7%. High Question/High Schema: SQL was not able to answer any question accurately. In comparison, SPARQL achieves an AOEA of 38.7%.

低问题/低表结构: SQL 是 25.5%, SPARQL 是 71.1%, 后者是前者的 2.8 倍。**高问题/低表结构**: SQL 是 37.4%, SPARQL 是 66.9%, 后者是前者的 1.8 倍。**低问题/高表结构**: SQL 一道题都没能准确回答, SPARQL 是 35.7%。**高问题/高表结构**: SQL 同样一道都没答对, SPARQL 是 38.7%。

Per the hypothesis, SPARQL achieves higher accuracy than SQL in every quadrant. Furthermore, it is surprising to observe that SQL was not able to answer any question in the High Schema Complexity quadrants. The results also lead us to understand when SQL starts to fail. When a question requires more than 5 tables to provide then answer, the accuracy drops to zero.

与假设一致, SPARQL 在每一个象限的准确率都高于 SQL。此外, 观察到 SQL 在高表结构复杂度的两个象限里一道题都答不对, 这一点令人意外。这些结果还让我们看清了 SQL 路线是从哪里开始失效的: **当一个问题需要连 5 张以上的表才能回答时, 准确率掉到零。**

5.3 部分正确

We consider a partially accurate answer to be one where the returned answers are accurate but incomplete. During the manual analysis, the following patterns for partially accurate answers are observed: Overlap: the columns returned by the query are correct, however, they are a subset of the accurate answer. Return Identifier: An internal identifier was returned instead of the appropriate label.

我们把「部分正确」定义为: 返回的答案是准确的, 但不完整。在人工分析中, 我们观察到部分正确的两种形态。**重叠**: 查询返回的列是对的, 但只是正确答案的一个子集, 有时还会包含一些不在预期答案里的列。**返回标识符**: 返回的是内部标识符, 而不是合适的业务标签。

From a practical perspective, we believe it is important to have a measure of partial accuracy because it connects the results to reality. In practice, if a user is interacting with a system and the results are missing a column, they could ask for the missing column or provide a label instead of an identifier. Therefore partial accuracy may be acceptable for users. However this is an open question on how to define partial accuracy and how to score it.

从实用角度看, 我们认为有一个衡量部分正确的尺度很重要, 因为它把结果和现实连了起来。实践中, 如果用户在和系统交互、结果少了一列, 他可以追问那一列, 或者要求给标签而不是标识符。因此**部分正确对用户来说可能是可接受的**。不过, 怎么定义部分正确、怎么给它打分, 仍是一个开放问题。

5.4 两种路线的错法完全不同

The following three types of inaccuracies were observed for SQL: Column Name Hallucinations: Column names were generated that do not exist in the corresponding table. Value Hallucinations: Generated value applied as a filter on a column where that value does not exist in the database. Join Hallucinations: Generated joins that are not accurate.

SQL 路线观察到三类错误。**列名幻觉**：生成了对应表里根本不存在的列名。**取值幻觉**：生成了一个取值当作某列的筛选条件，而这个值在数据库里并不存在。**连接幻觉**：生成了不正确的表连接。

For SPARQL: Incorrect Path: The generated query does not follow the correct path of the properties in the ontology. The generated path goes from A to C when the correct path is A to B to C. Incorrect Direction: The generated query swaps the direction of a property. The generated direction is B to A, when the correct direction is A to B.

SPARQL 路线则是另外两类。**路径错误**：生成的查询没有沿着本体里正确的属性路径走，正确路径是从 A 经 B 到 C，它却直接从 A 到 C。**方向错误**：生成的查询把某个属性的方向搞反了，正确方向是从 A 到 B，它写成了从 B 到 A。

Discussion The inaccuracy of SQL queries are based on hallucination while the inaccuracy of SPARQL queries are based on path inconsistency. To our surprise, while manually analyzing the generated SPARQL queries, we did not observe a hallucinated class or property.

讨论：**SQL 查询的错误源于幻觉，而 SPARQL 查询的错误源于路径不一致。** 令我们意外的是，在人工分析生成的 SPARQL 查询时，**我们没有观察到任何一个被编造出来的类或属性。**

The SQL hallucinations are evident: column names that don't exist in a table and values that the LLM does not know if they exist. On the wrong joins, one can argue that the joins have been hallucinated because at an initial glance, the joins may seem plausible, but they are not how the database was designed, thus they have been made up by the LLM.

SQL 那边的幻觉是明摆着的：表里不存在的列名，以及大模型并不知道是否存在的取值。至于错误的连接，可以说那些连接也是幻觉出来的——**乍一看它们似乎讲得通，但那并不是这个数据库被设计出来的样子，所以是模型编的。**

For SPARQL queries, one can also argue that the paths were hallucinated because they are incorrect. However, the generated paths were indicative that the LLM knew what the correct starting and end node was and the miss was on defining the correct path from the start node to the end node. One could even argue that the LLM appears to do some sort of reasoning but not always getting it correct.

对 SPARQL 查询，也可以说那些路径是幻觉，因为它们确实不对。不过，生成的路径表明**大模型知道正确的起点和终点是什么，失手的地方在于确定从起点到终点的那条正确路径。** 甚至可以说，模型看上去在做某种推理，只是不总能做对。

六、结论：语境必须被当作一等公民

What is evident is that context is crucial for accuracy and these results further emphasizes the need to invest in context. The point to be made here is a call to action that investing in context of SQL databases is required to increase the accuracy of LLMs for question answering over the SQL databases.

显而易见的是，**语境对准确率至关重要**，这些结果进一步强调了投资于语境的必要性。这里要说的是一个行动号召：要提升大模型在 SQL 数据库上问答的准确率，就必须投资于这些数据库的语境。

In the benchmark, the context was presented in the form of an ontology that describes the semantics of the business domain and mappings that connect the physical schema with the ontology which are used to create the knowledge graph. The ontology can also include further semantics such as synonyms, labels in different languages, which are not expressible in a SQL DDL.

在这个基准里，语境以两种形式呈现：一份描述业务领域语义的本体，以及一套把物理表结构与本体连接起来的映射，两者共同用来构建知识图谱。**本体还可以包含更多语义，比如同义词、不同语言的标签，这些都是 SQL DDL 表达不了的。**

We argue that this context needs to be treated as a first class citizen, thus, at minimum managed in a metadata management system (i.e. data catalog) and ideally on a knowledge graph architecture. Otherwise, the crucial context that provides accuracy would be managed in an ad-hoc manner.

我们主张这层语境必须被当作一等公民对待，至少要在元数据管理系统（也就是数据目录）里管起来，理想情况下管在知识图谱架构上。**否则，这份提供准确率的关键语境就会被以临时凑合的方式管理。**

七、研究议程：作者自己列的待办

Generative AI and Large Language Models are following the traditional hype cycle. It is rapidly reaching the peak of inflated expectations. We posit that applying these technologies to enterprise data will follow with the trough of disillusionment due to the lack of accuracy, explainability and governance.

生成式 AI 与大语言模型正沿着传统的技术炒作周期走，它正迅速逼近「期望膨胀的顶峰」。我们认为，把这些技术应用到企业数据上，接下来会因为**准确率、可解释性与治理**这三样的缺失而跌入「幻灭的低谷」。

Knowledge Engineering Methodologies: A finding of our work is the need to investing in context of SQL databases in the form of ontology and mappings which define the knowledge graph. Existing ontology and knowledge engineering methodologies have been defined in an era of smaller amount of data, and are manual processes. How can these methodologies be adapted to an era of large amounts of data and even incorporating LLMs in order increase productivity?

知识工程方法论：我们工作的一个发现是，需要以本体与映射的形式投资于 SQL 数据库的语境，这两者共同定义了知识图谱。**现有的本体与知识工程方法论是在数据量较小的年代定下的，而且都是手工流程。**这些方法论要怎么适应大数据量的年代，甚至怎么把大模型也纳进来以提升效率？

Business Implications: What would accurate enough mean? Enterprise expect perfect results, but how do they know that their current approaches are accurate enough? These LLM-powered question answering systems should be compared to existing (manual) approaches that answer questions (i.e. the steps to answer a question by reviewing a dashboard).

业务含义：「足够准确」到底是什么意思？企业期待完美的结果，**但他们怎么知道自己现有的做法就足够准确呢？**这些大模型驱动的问答系统，应该拿去和现有的（人工的）回答方式作比较，比如「靠翻看一个看板来回答一个问题」的那些步骤。

八、总结

The results of benchmark determined that using GPT-4 and zero-shot prompting, Enterprise Natural Language questions over Enterprise SQL databases achieved 16.7% accuracy, which increased to 54.2% when a Knowledge Graph representation was used. The main conclusion of this work is that investing in Knowledge Graphs provides higher accuracy for LLM powered question answering systems.

基准的结果确定了：使用 GPT-4 和零样本提示，企业自然语言问题在企业 SQL 数据库上的准确率是 16.7%；当改用知识图谱表示时，这个数字上升到 54.2%。**本工作的主要结论是：投资于知识图谱，能让大模型驱动的问答系统获得更高的准确率。**

参考链接

- **基准的全部数据与代码** — 作者公开的仓库，含建表语句、样例数据、OWL 本体、R2RML 映射、43 道问题与参考查询，可以拿自己公司的表结构重跑
- **OMG 财产与意外险数据模型** — 基准所用的企业级标准表结构，物理模型 199 张表、243 个外键
- **Spider 基准** — 本文的执行准确率指标沿用自它，是 Text-to-SQL 领域最常用的公开基准之一
- **大模型与知识图谱结合的资源汇总** — 本文引用的这个方向的资料索引
- **HELM 语言模型整体评估** — 斯坦福的模型评估框架，作者把「按 HELM 的路子做」列进了后续计划